

Name	Subject Class	Class	Candidate Number
	2BIX01		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2011

BIOLOGY

HIGHER 1

Paper 2

8875/02
15 AUGUST 2011
2 hours

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.
Fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
5 / 6	
Total	60

This question paper consists of **10** printed pages.

- 1 (a) Fig. 1.1 below is an electron micrograph showing part of an animal cell.

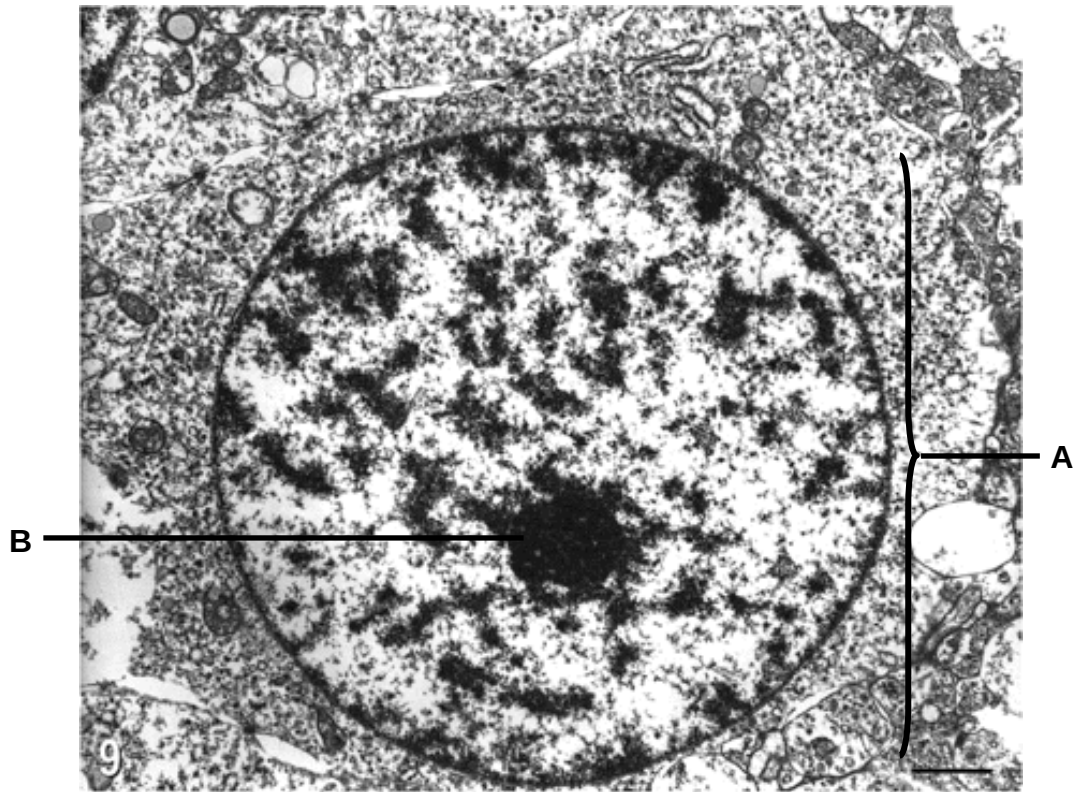


Fig. 1.1

- (i) Name the structure labelled **B**.

B: _____ [1]

- (ii) State **one** visible feature that enabled your identification of structure **B** and describe the role of structure **B**.

 _____ [2]

- (iii) Organelle **A** contains threadlike materials that are made up of protein and non-protein. These materials are crucial to the survival and propagation of cells. Describe the structure of the non-protein material.

 _____ [3]

(iv) Describe the roles of rRNAs in protein synthesis.

[2]

(v) Briefly describe the process of transcription.

[3]

(b) Human Genome Project was set up to characterise the genomes of humans and other organisms. Discuss **three** ethical concerns of the Human Genome Project.

[3]

[Total: 14]

- 2 In a mixture containing oil and water, phospholipids will arrange themselves to form a monolayer at the interface between oil and water as shown in Fig. 2.1.

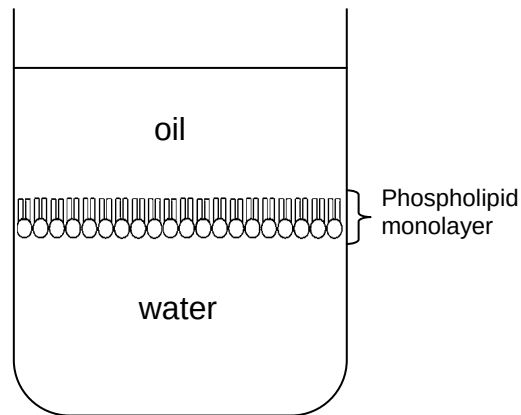


Fig. 2.1

- (a) State **two** structural differences between a glucose molecule and a phospholipid.

[2]

- (b) With reference to Fig. 2.1, explain why the phospholipid molecules form the monolayer as shown.

[2]

- (c) (i) Explain why the term “fluid mosaic model” is used to describe the plasma membrane.

[1]

- (ii) Describe how large insoluble particles may be taken into a cell.

[2]

- (d) Some detergents have very similar structures to phospholipids. Suggest how they can be used to extract membrane proteins from a cell.

[1]

[Total: 8]

- 3 (a) Tail length in cats is determined by the Japanese Bobtail gene. The normal wild-type allele **T** is dominant and codes for normal tail length in cats. The mutant allele **t** is recessive and codes for a short tail, as found in the Japanese Bobtail breed, hence the name.

Coat colour in cats is determined by the *tyr* gene, which codes for tyrosinase enzyme. This enzyme is involved in pigment production, and controls the intensity of body colour in cats. There are five different alleles for the *tyr* gene, namely full colour **B**, Burmese **b^M**, Siamese **b^S**, blue-eyed albino **b^A**, and albino **b**. Each allele gives rise to a different mutant tyrosinase enzyme, resulting in the different phenotypes observed.

Table 3.1 below shows the different breeds possible.

Table 3.1

	B	b^M	b^S	b^A	b
B	Normal	Normal	Normal	Normal	Normal
b^M		Burmese	Tonkinese	Burmese	Burmese
b^S			Siamese	Siamese	Siamese
b^A				Blue-Eyed Albino	Blue-Eyed Albino
b					Albino

- (i) With reference to Table 3.1 above, determine the relative dominance of the alleles **B**, **b^M** and **b^S**.

[1]

- (ii) A homozygous Burmese female cat with normal tail was crossed with a homozygous Siamese male cat with a short tail. The F_1 offspring were then sibling-mated to form the F_2 generation. Draw a genetic diagram to show the expected genotypes and phenotypes of the F_2 offspring.

[4]

- (iii) A cat owner wanted to find out if his Siamese female cat was a pure-bred. Suggest how he could do so.

[2]

- (b) Tortoiseshell cats are usually female, with a combination of black and orange fur. During skin cell development in the foetal stage, one of the X chromosomes in every cell will be randomly inactivated. The allele on the remaining X chromosome is expressed. The allele for orange fur (X^O) is codominant to the allele for black fur (X^B).

- (i) State the probability of producing a female tortoiseshell kitten, when a female tortoiseshell cat mates with a black male cat.

[1]

- (ii) Approximately 1 in 3,000 tortoiseshell cats are male. Based on your knowledge of meiosis, suggest how male tortoiseshell cats may be produced.

[2]

[Total: 10]

- 4 Magneto is a recently discovered sex-linked disease. It is caused by mutation at two nucleotide positions which coincide with the restriction site of restriction enzyme *EcoR1*. Fig. 4.1 shows the pedigree diagram of the sex-linked disease. Fig. 4.2 shows the two alleles obtained from the DNA of individuals in the family. Positions of *EcoR1* restriction sites are marked by arrows.

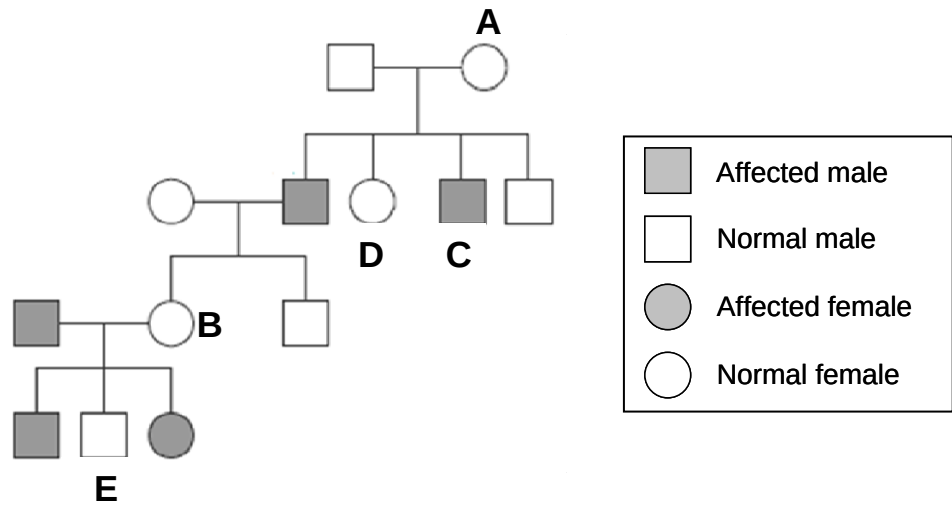


Fig. 4.1

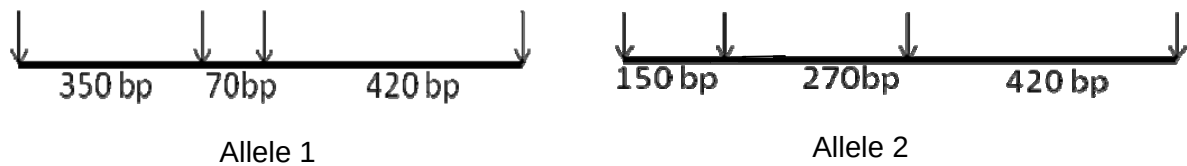


Fig. 4.2

To determine the allele causing the sex-linked disease, DNA analysis was carried out on individuals A, B, C, D and E. Fig. 4.3 shows a gel electropherogram obtained from the DNA analysis of the alleles from individuals A and D.

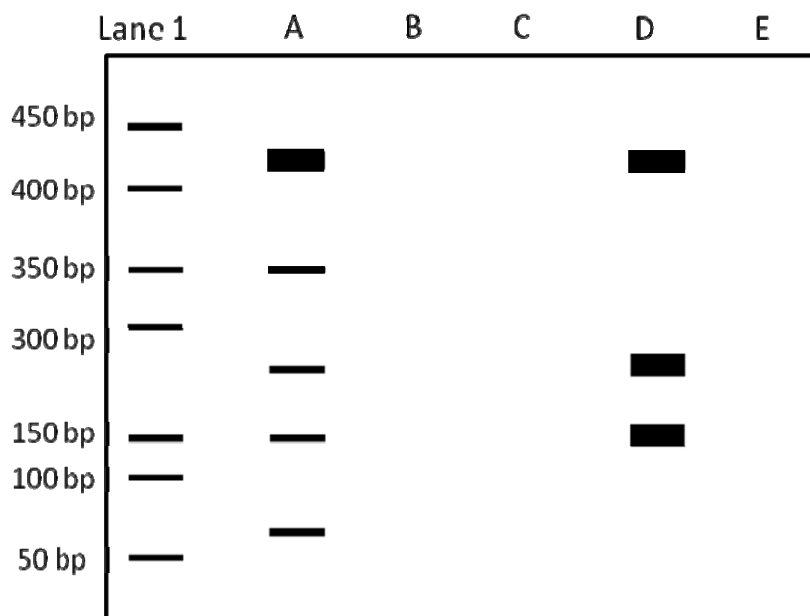


Fig. 4.3

- (a) (i)** With reference to Fig. 4.1, Fig. 4.2, and the band patterns of individuals **A** and **D** in Fig. 4.3, identify with reasons the disease-causing allele.

[3]

- (ii)** Using Fig. 4.2, suggest how the disease-causing allele was formed by mutation at two nucleotide positions.

[1]

- (iii)** Draw on Fig. 4.3 the band patterns of individuals **B**, **C** and **E**.

[3]

- (b)** Explain briefly how the band patterns are visualised.

[1]

[Total: 8]

Section B

Answer **one** question.

Write your answers in the lined pages provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Compare and contrast photophosphorylation and oxidative phosphorylation. [9]
- (b)** Assuming that there is no genetic variation and no migration into and out of a population, explain how evolution would be affected in this population. [6]
- (c)** Using **one** named example, explain the role of the environment in evolution. [5]
- 6 (a)** Explain how the structure of the following proteins is adapted to their functions:
- (i)** Collagen; [5]
- (ii)** Haemoglobin. [6]
- (b)** Assuming that the environment remains constant and there is no migration into and out of a population, explain why the members of this population could be phenotypically different. [9]